

My Work sample

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Introduction

This project analyzes the relationship between income (GDP) and health outcomes (life expectancy) across countries.

Data

The dataset includes country-level information on GDP and life expectancy over time. These variables allow us to examine how income is associated with health outcomes across different countries.

1. Data Introduction

```
# 1. Load library
library(tidyverse)

# 2. Read dataset
my_data <- read.csv("life_expectancy_gdp.csv")

# 3. Check dataset
summary(my_data)
```

```
##      Country      Year  Life expectancy at birth..years.
## Length:96      Min.   :2000      Min.   :44.30
## Class :character 1st Qu.:2004      1st Qu.:74.47
## Mode  :character Median :2008      Median :76.75
##              Mean   :2008      Mean   :72.79
##              3rd Qu.:2011      3rd Qu.:78.90
##              Max.   :2015      Max.   :81.00
##      GDP
## Min.   :4.416e+09
## 1st Qu.:1.733e+11
## Median :1.280e+12
## Mean   :3.880e+12
## 3rd Qu.:4.068e+12
## Max.   :1.810e+13
```

```
# 4. Clean dataset
data_clean <- my_data %>%
  select(
    country = Country,
```

```

year = Year,
life_exp = Life.expectancy.at.birth..years.,
gdp = GDP
)

```

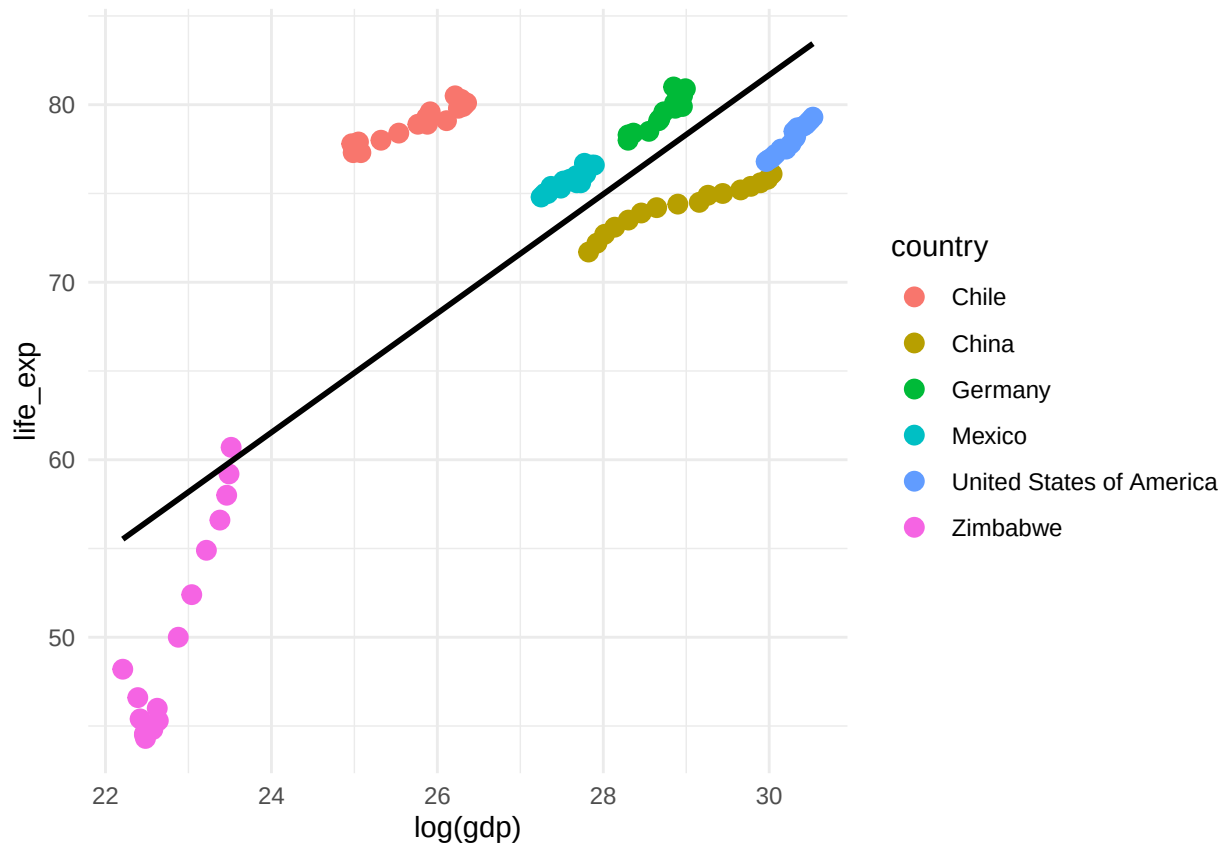
2. Descriptive Visualization

The following chart explores the relationship between GDP and Life Expectancy.

```

# drawing plot
ggplot(data_clean, aes(x = log(gdp), y = life_exp, color = country)) +
  geom_point(size = 3) +
  geom_smooth(method = "lm", color = "black", se = FALSE) +
  theme_minimal()

```



The plot shows a clear positive relationship between GDP and life expectancy. Countries with higher income levels tend to have higher life expectancy. Additionally, within each country, both GDP and life expectancy increase over time.

3. Statistical Analysis

To quantify the relationship, I estimate a simple linear model.

```

# running regression
model <- lm(life_exp ~ log(gdp), data = data_clean)

# show the result
summary(model)

##
## Call:
## lm(formula = life_exp ~ log(gdp), data = data_clean)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -12.145  -4.610   1.171   2.354  13.013
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -19.001      7.444  -2.553  0.0123 *
## log(gdp)       3.356      0.271  12.382 <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6.615 on 94 degrees of freedom
## Multiple R-squared:  0.6199, Adjusted R-squared:  0.6159
## F-statistic: 153.3 on 1 and 94 DF,  p-value: < 2.2e-16

```

The regression results show a strong positive relationship between GDP and life expectancy. The coefficient on $\log(\text{GDP})$ is positive and highly statistically significant ($p < 0.001$), indicating that higher income levels are associated with longer life expectancy. The model explains a substantial share of the variation in life expectancy (R^2 is about 0.62).